

Given the following sentence construct the probability table:

1. <s> the system processes the text<s>
2. <s>the model generates a text<s>
3. <s>a system processes the model<s>
4. <s>the model learns a pattern<s>
5. <s>a text generates the pattern<s>
6. <s>the system learns a text<s>
7. <s>a pattern processes the text<s>

Word	Count
<s> start	7
the	8
system	3
processes	3
text	5
model	3
generates	2
a	6
learns	2
pattern	3
<s> end	7

Vocabulary Length = 10 Note <s> counts as 1 element

Count Table

	<s>	the	system	processes	text	model	generates	a	learns	pattern
<s>	0	4	0	0	0	0	0	3	0	0
the	0	0	2	0	2	3	0	0	0	1
system	0	0	0	2	0	0	0	0	1	0
processes	0	3/3	0	0	0	0	0	0	0	0
text	4	0	0	0	0	0	1	0	0	0
model	1	0	0	0	0	0	1	0	1	0
generates	0	1	0	0	0	0	0	1	0	0
a	0	0	1	0	3	0	0	0	0	1
learns	0	0	0	0	0	0	0	2	0	0
pattern	2	0	0	1	0	0	0	0	0	0

Probability Table

	<s>	the	system	processes	text	model	generates	a	learns	pattern
<s>	0	4/7	0	0	0	0	0	3/7	0	0
the	0	0	2/8	0	2/8	3/8	0	0	0	1/8
system	0	0	0	2/3	0	0	0	0	1/3	0
processes	0	3/3	0	0	0	0	0	0	0	0
text	4/5	0	0	0	0	0	1/5	0	0	0
model	1/3	0	0	0	0	0	1/3	0	1/3	0
generates	0	1/2	0	0	0	0	0	1/2	0	0
a	0	0	1/6	0	3/6	0	0	0	0	2/6
learns	0	0	0	0	0	0	0	2/2	0	0
pattern	2/3	0	0	1/3	0	0	0	0	0	0

Calculate the following:

$P(<s>\text{the system processes the text}<s>)$:

$$P(\text{the}|<s>) = 4/7$$

$$P(\text{system}|\text{the}) = 2/8$$

$$P(\text{processes}|\text{system}) = 2/3$$

$$P(\text{the}|\text{processes}) = 1$$

$$P(\text{text}|\text{the}) = 2/8$$

$$P(<s>|\text{text}) = 4/5$$

$$\text{Answer} = 0.018$$

$P(<s>\text{a text generates the pattern}<s>)$

$$P(\text{a}|<s>) = 3/7$$

$$P(\text{text}|\text{a}) = 3/6$$

$$P(\text{generates}|\text{text}) = 1/5$$

$$P(\text{the}|\text{generates}) = 1/2$$

$$P(\text{pattern}|\text{the}) = 1/8$$

$$P(<s>|\text{pattern}) = 2/3$$

$$\text{Answer} = 0.001785$$

$P(<s>\text{the pattern learns a model}<s>)$

$$P(\text{the}|<s>) = 4/7$$

$$P(\text{pattern}|\text{the}) = 1/8$$

$$P(\text{learns}|\text{pattern}) = 0$$

$$P(\text{a}|\text{learns}) = 1$$

$$P(\text{model}|\text{a}) = 0$$

$$P(<s>|\text{model}) = 1/3$$

$$\text{Answer} = 0$$

Applying Add-1”Laplace Probability Equation”

$$P(W_i | W_{(i-1)}) = (\text{Count}(W_{(i-1)}, W_i) + 1) / (\text{Count}(W_{(i-1)}) + 10(V))$$

	<s>	the	system	processes	text	model	generates	a	learns	pattern
<s>	1/17	5/17	1/17	1/17	1/17	1/17	1/17	4/17	1/17	1/17
the	1/18	1/18	3/18	1/18	3/18	4/18	1/18	1/18	1/18	2/18
system	1/13	1/13	1/13	3/13	1/13	1/13	1/13	1/13	2/13	1/13
processes	1/13	4/13	1/13	1/13	1/13	1/13	1/13	1/13	1/13	1/13
text	5/15	1/15	1/15	1/15	1/15	1/15	2/15	1/15	1/15	1/15
model	2/13	1/13	1/13	1/13	1/13	1/13	2/13	1/13	2/13	1/13
generates	1/12	2/12	1/12	1/12	1/12	1/12	1/12	2/12	1/12	1/12
a	1/16	1/16	2/16	1/16	4/16	1/16	1/16	1/16	1/16	3/16
learns	1/12	1/12	1/12	1/12	1/12	1/12	1/12	3/12	1/12	1/12
pattern	3/13	1/13	1/13	2/13	1/13	1/13	1/13	1/13	1/13	1/13

P(<s>the pattern learns a model<s>)

P(the|<s>) = 5/17

P(pattern|the) = 2/18

P(learns|pattern) = 1/13

P(a|learns) = 3/12

P(model|a) = 1/16

P(<s>|model) = 2/13

Answer = 0.000006